DOMAIN 3: AI-BASED CYBERSECURITY

Focus: Intelligent threat detection, social engineering prevention, network security, digital forensics, and adaptive cyber defense using artificial intelligence and machine learning.

ABCS-01: Intelligent Cyber Threat Detection

Problem Statement

As cyberattacks continue to evolve, traditional security mechanisms often struggle to identify previously unseen threats, sophisticated attack patterns, and suspicious digital activities. High volumes of security alerts and false positives make it difficult for security teams to distinguish genuine threats from normal system behavior.

Challenge: Develop an AI-powered cyber threat detection system that analyzes system logs, user activities, and security events to identify suspicious behavior, detect emerging threats, and assess potential risks. The system should prioritize security alerts, minimize false positives, and provide actionable insights to help security teams respond effectively.

Expected Solution

- AI-Based Threat Detection: Identify malicious activities using machine learning and behavioral analysis.
- Anomaly Detection: Detect unusual system behavior and previously unseen attack patterns.
- Threat Risk Assessment: Classify detected threats according to their severity and potential impact.
- Real-Time Security Alerts: Notify security teams about suspicious activities and potential attacks.
- Security Dashboard: Visualize detected threats, attack trends, risk levels, and incident history.

Suggested Technologies: Python, Scikit-learn, TensorFlow or PyTorch, Pandas, SIEM datasets, FastAPI, React.js, PostgreSQL.

Impact: Earlier threat detection, reduced security risks, improved incident response, and stronger protection against emerging cyberattacks.

ABCS-02: AI-Assisted Digital Forensics

Problem Statement

The growing volume and complexity of digital evidence make cybercrime investigations increasingly challenging. Security investigators must examine system logs, files, network records, and other digital artifacts to identify relevant evidence, reconstruct event sequences, discover hidden patterns, and investigate potential evidence manipulation. These processes can be time-consuming and require specialized expertise.

Challenge: Develop an AI-assisted digital forensics platform that helps investigators analyze digital artifacts, correlate security events, identify suspicious patterns, and reconstruct potential incident timelines. The system should generate evidence summaries, highlight relevant findings, and maintain evidence integrity through secure handling, cryptographic hashing, and auditable investigation records.

Expected Solution

- Digital Evidence Analysis: Process system logs, file metadata, and authorized digital artifacts to identify relevant information.
- AI-Based Pattern Discovery: Detect relationships between events, suspicious activities, and potential attack indicators.
- Incident Timeline Reconstruction: Organize evidence chronologically to help investigators understand how an incident may have occurred.
- Evidence Integrity Verification: Use cryptographic hashes, such as SHA-256, to verify that collected evidence has not changed.
- Automated Investigation Reports: Generate structured summaries containing findings, timestamps, and references to supporting evidence.
- Investigation Dashboard: Enable authorized investigators to review evidence, track analysis progress, and maintain an audit trail.

Suggested Technologies: Python, Pandas, Scikit-learn, NLP, YARA, SQLite or PostgreSQL, SHA-256 hashing, FastAPI, React.js.

Impact: Faster digital investigations, improved evidence organization, more reliable incident analysis, and stronger accountability in cybercrime investigations.

ABCS-03: Adaptive & Intelligent Cyber Defense

Problem Statement

Organizations face increasing difficulties in prioritizing security alerts, responding to emerging threats, and minimizing the impact of cyber incidents. Conventional security approaches may rely on static rules and manual intervention, limiting their ability to adapt to changing attack patterns and rapidly evolving threats. Poorly prioritized alerts can delay critical responses and overwhelm security teams.

Challenge: Develop an AI-assisted adaptive cyber defense platform that correlates security alerts, evaluates threat severity, recommends response strategies, and learns from historical incident patterns. The system should prioritize critical events, support coordinated incident response, and recommend defensive actions while maintaining human oversight and appropriate authorization for high-impact operations.

Expected Solution

- Intelligent Alert Prioritization: Rank security alerts based on severity, confidence, asset importance, and potential impact.
- Threat Correlation: Combine related security events to identify broader attack patterns and possible attack chains.
- Adaptive Risk Assessment: Update threat assessments as new evidence and security events become available.
- Automated Response Recommendations: Suggest appropriate defensive actions, such as isolating a test endpoint or disabling a compromised test account, subject to authorization.
- Incident Response Dashboard: Track alerts, recommended actions, approvals, and incident resolution progress.
- Human-in-the-Loop Control: Require authorized human approval for high-impact actions and maintain an auditable record of decisions.

Suggested Technologies: Python, Scikit-learn, XGBoost, PyTorch, FastAPI, SIEM data, MITRE ATT&CK framework, React.js, PostgreSQL, workflow automation tools.

Impact: Faster incident response, better alert prioritization, reduced security team workload, and more adaptive and reliable cyber defense.